

Explanation of Loss Functions: MCC, Focal Loss, and Dice Loss

Note on Current Model

The current implementation uses:

```
loss_fn = nn.CrossEntropyLoss(weight=weights)
```

This implies:

- The model is optimized using Cross Entropy Loss.
- Class imbalance is handled using class weighting (`weights`).

1 MCC Loss (Matthews Correlation Coefficient Loss)

MCC is a robust metric for evaluating classification performance, especially in imbalanced datasets.

1.1 Formula

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Where:

- TP: True Positive
- TN: True Negative
- FP: False Positive
- FN: False Negative

1.2 Loss Function

$$Loss = 1 - MCC$$

1.3 Range and Interpretation

- $-1 \leq MCC \leq 1$
- $0 \leq Loss \leq 2$

Interpretation:

- Perfect model: $MCC = 1 \Rightarrow Loss = 0$
- Random model: $MCC \approx 0 \Rightarrow Loss \approx 1$
- Worst model: $MCC = -1 \Rightarrow Loss = 2$

1.4 When to Use

- Highly imbalanced classification
- Medical diagnosis problems
- Rare event detection

2 Focal Loss

Focal Loss improves Cross Entropy by focusing on hard-to-classify samples.

2.1 Formula

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t)$$

Where:

- p_t : predicted probability of the correct class
- α : class balancing factor
- γ : focusing parameter

2.2 Hyperparameters

Alpha (α):

- Typical range: $0.1 - 0.9$
- Common value: 0.25

Gamma (γ):

- Typical range: $1 - 5$
- Common value: 2

2.3 Behavior

- $\gamma = 0$: Equivalent to Cross Entropy Loss
- Higher γ : Focuses more on hard examples

2.4 Practical Loss Range

- Typically: $0 - 5$
- Good training: < 0.5

3 Dice Loss

Dice Loss is commonly used in segmentation tasks to measure overlap.

3.1 Dice Coefficient

$$Dice = \frac{2|X \cap Y|}{|X| + |Y|}$$

3.2 Soft Dice Formulation

$$Dice = \frac{2 \sum (p \cdot y) + \epsilon}{\sum p + \sum y + \epsilon}$$

3.3 Loss Function

$$DiceLoss = 1 - Dice$$

3.4 Range and Interpretation

- $0 \leq Dice \leq 1$
- $0 \leq DiceLoss \leq 1$

Interpretation:

- Perfect overlap: Loss = 0
- No overlap: Loss = 1

4 Comparison Summary

Loss Function	Range	Best Value
MCC Loss	0 – 2	0
Focal Loss	0 – ∞ (practical 0–5)	0
Dice Loss	0 – 1	0

5 Recommended Hyperparameters

Focal Loss (recommended starting point):

- $\alpha = 0.25$
- $\gamma = 2$

Tversky/Dice variant (segmentation):

- $\alpha = 0.3$
- $\beta = 0.7$

6 Final Recommendation

For image classification with imbalance:

- Use Cross Entropy with weights (baseline)
- Try Focal Loss for better imbalance handling
- Use MCC Loss for extreme imbalance
- Use Dice Loss for segmentation tasks